

Hardware Components and Functional Description

Project Title

AI-Enabled Low-Cost Real-Time Mine Subsidence Monitoring, Prediction and Early Warning System for Underground Coal Mines in India

Hardware System Overview

The proposed system consists of a distributed network of low-cost sensor nodes installed on the surface above underground coal mining areas. Each sensor node continuously measures parameters such as ground inclination, vibration, displacement, and crack development. The collected data is transmitted through a LoRa-based wireless network to a Raspberry Pi gateway for local processing, storage, and communication with the AI and monitoring platform.

The hardware architecture is designed to be **low-cost, low-power, scalable, and suitable for continuous outdoor monitoring.**

1. Heltec WiFi LoRa 32 V3

Component

Heltec WiFi LoRa 32 V3

Main Function

Acts as the **main controller and communication unit** of each sensor node.

Purpose in the Project

The Heltec board collects data from all the sensors connected to the node, performs initial processing, and transmits the sensor data through LoRa.

Functions

- Reads sensor measurements.
- Processes sensor data using the ESP32 microcontroller.
- Calculates/handles tilt and acceleration measurements.
- Controls connected sensors.
- Transmits data through LoRa.
- Supports wireless communication between nodes/gateway.
- Monitors battery status.

- Implements deep-sleep operation for power saving.

Why It Is Used

The Heltec WiFi LoRa 32 V3 combines an **ESP32 microcontroller and LoRa communication** in a single board. This reduces the number of separate components required and makes the sensor node compact and cost-effective.

Project Role

Main controller + LoRa communication + local sensor processing

2. LoRa Wireless Communication

Component

LoRa Module / LoRa built into Heltec WiFi LoRa 32 V3

Main Function

Provides **long-range, low-power wireless communication** between distributed sensor nodes and the gateway.

Purpose in the Project

Sensor nodes may be distributed over a large surface area above underground mine panels. LoRa allows the sensor nodes to transmit small amounts of monitoring data over relatively long distances while consuming comparatively little power.

Functions

- Transmits sensor data wirelessly.
- Receives data from other nodes where applicable.
- Supports distributed sensor networking.
- Enables communication between sensor nodes and gateway.
- Reduces dependence on wired communication.

Example Data

Node ID : N03

Tilt : 1.24°

Vibration : 0.18 g

Displacement : 2.4 mm

Battery : 82%

Status : WARNING

Project Role

Wireless data transmission for the surface sensor network

Important Innovation

The project can implement **multi-node/multi-hop communication** so that sensor nodes can forward data through neighboring nodes when direct communication with the gateway is unavailable. This supports the project's **Wireless Surface Mesh Network** concept.

3. MPU6050

Component

MPU6050 Accelerometer and Gyroscope

Main Function

Measures **inclination/tilt and motion** of the sensor node.

Purpose in the Project

Ground deformation may cause small changes in the orientation of a surface-mounted sensor node. The MPU6050 detects changes in orientation and acceleration that can indicate abnormal ground movement.

Functions

- Measures acceleration along three axes.
- Measures angular velocity along three axes.
- Detects changes in inclination.
- Provides motion-related data.
- Helps identify gradual ground deformation.

Example

Normal condition:

Tilt = 0.2°

Ground movement:

Tilt = 1.1°

Further deformation:

Tilt = 2.0°

These values are only illustrative; actual warning thresholds must be determined through calibration and geotechnical validation.

Project Role

Ground tilt/inclination monitoring

4. ADXL345

Component

ADXL345 3-Axis Digital Accelerometer

Main Function

Measures **acceleration and vibration characteristics**.

Purpose in the Project

The ADXL345 provides numerical acceleration measurements that can be processed to identify unusual vibration patterns associated with ground movement or other disturbances.

Functions

- Measures acceleration in X, Y and Z axes.
- Detects changes in acceleration.
- Captures vibration-related data.
- Provides numerical sensor readings for analysis.
- Supports feature extraction for AI/ML processing.

Why We Prefer It for the Main Vibration Measurement

Unlike a simple vibration switch that mainly provides an ON/OFF indication, the ADXL345 provides **quantitative acceleration data**.

This is more useful for:

- anomaly detection
- vibration pattern analysis
- historical data analysis
- AI-based prediction

Project Role

Quantitative vibration and acceleration monitoring

5. SW-420 Vibration Sensor — Optional

Component

SW-420 Vibration Sensor

Main Function

Detects whether vibration is present.

Purpose in the Project

The SW-420 can be used as an inexpensive additional vibration trigger during prototype development.

Functions

- Detects vibration.
- Provides a simple digital trigger.
- Can activate an alert when vibration exceeds its configured sensitivity.

Limitation

It provides relatively basic vibration detection compared with the ADXL345.

Therefore:

ADXL345 → Primary vibration measurement

SW-420 → Optional low-cost vibration trigger

Project Role

Additional low-cost vibration detection

6. Linear Displacement Sensor

Component

Linear Displacement Sensor / Linear Potentiometer

Main Function

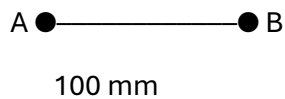
Measures **relative ground displacement or change in distance**.

Purpose in the Project

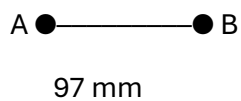
Subsidence can cause relative movement between two surface points. A displacement sensor can detect changes in distance that may indicate ground deformation.

Example

Initial position



After deformation



The system detects the change in distance.

Functions

- Measures linear movement.
- Detects relative displacement.
- Converts physical movement into an electrical signal.
- Provides deformation data to the ESP32.

Project Role

Ground stretch/displacement monitoring

Prototype Note

A linear potentiometer can be used for a low-cost demonstration. For field deployment, a calibrated displacement transducer would be preferable.

7. Crack Detection Sensor

Component

Crack Sensor / Crack-Width Measurement Mechanism

Main Function

Detects **crack initiation or crack-width changes**.

Purpose in the Project

Surface cracks can be an important visible indication of ground deformation. The sensor is intended to identify the formation or widening of cracks.

Functions

- Detects crack opening.
- Measures changes in crack width where supported by the chosen sensor arrangement.
- Provides early deformation information.
- Sends crack-related data to the controller.

Project Role

Early crack detection

8. 18650 Li-ion Battery

Component

18650 Lithium-ion Battery — approximately 3000 mAh

Main Function

Provides electrical power to the sensor node.

Purpose in the Project

The sensor nodes are intended to operate continuously in outdoor locations where direct electrical power may not be available.

Functions

- Powers the ESP32.
- Powers sensors.
- Powers LoRa communication.
- Provides backup power during low-sunlight conditions.

Project Role

Primary energy storage

9. Solar Panel

Component

Small Solar Panel

Main Function

Provides renewable energy for charging the battery.

Purpose in the Project

Because the sensor nodes may be deployed for long periods, solar charging can extend operating time and reduce the need for frequent battery replacement.

Functions

- Converts sunlight into electrical energy.
- Charges the battery through appropriate charging/power-management circuitry.
- Supports long-term autonomous operation.

Project Role

Renewable power source for long-term deployment

10. Battery Charging and Protection Circuit

Components

- **TP4056-based Li-ion charging circuit**
- **DW01 battery protection**, where applicable

Main Function

Provides battery charging and protection functions.

Purpose in the Project

The rechargeable battery must be safely managed during charging and operation.

Functions

- Controls Li-ion battery charging.
- Provides battery protection when using an appropriate protected configuration.

- Helps prevent unsafe battery operating conditions.

Important Design Note

For the final outdoor solar-powered node, the charging circuit should be selected/configured specifically for the solar panel, battery chemistry, and required protection features. A basic TP4056 module should not automatically be treated as a complete solar power-management solution.

Project Role

Battery charging and protection

11. ESP32 Deep Sleep

Feature

ESP32 Deep-Sleep Power Management

Main Function

Reduces power consumption when continuous sensor processing/transmission is not required.

Purpose in the Project

The sensor node is battery powered. Keeping the ESP32 fully active continuously would consume more energy.

Instead:

Wake Up

↓

Read Sensors

↓

Process Data

↓

Transmit Through LoRa

↓

Sleep

↓

Wake Up Again

Functions

- Reduces average power consumption.
- Extends battery operating time.
- Allows periodic sensor measurements.
- Supports autonomous operation.

Project Role

Low-power operation and extended battery life

12. Raspberry Pi Gateway

Component

Raspberry Pi

Main Function

Acts as the **central gateway and edge-processing unit** for the sensor network.

Purpose in the Project

Instead of using a Raspberry Pi for every sensor node, one Raspberry Pi can collect data from multiple LoRa sensor nodes.

Functions

- Receives LoRa sensor data.
- Collects data from multiple sensor nodes.
- Stores sensor data locally.
- Performs preliminary/edge data processing.
- Can run lightweight anomaly-detection models.
- Provides local monitoring capability.
- Synchronizes data with a cloud/server when connectivity is available.
- Can activate local warning devices.

Example

Sensor Nodes

↓

LoRa Network

↓

Raspberry Pi

↓

Local Database

↓

Edge Processing

↓

Cloud / AI Platform

Project Role

LoRa gateway + local processing + data storage + edge computing

13. Buzzer / Local Warning Unit

Component

Buzzer / Siren + LED indicators

Main Function

Provides a **local warning when abnormal conditions are detected.**

Purpose in the Project

The system should not depend completely on internet connectivity for local alerts.

Example

GREEN → Normal

YELLOW → Warning

RED → Critical

The Raspberry Pi/gateway or appropriate controller can activate the local warning mechanism based on validated prototype risk logic.

Project Role

Local early-warning indication

14. Waterproof Enclosure

Component

Weather-resistant enclosure

Main Function

Protects electronics from environmental conditions.

Purpose in the Project

Surface sensor nodes may be exposed to:

- rain
- dust
- moisture
- sunlight
- temperature variation

Functions

- Protects the ESP32.
- Protects sensors and wiring.
- Improves reliability.
- Allows outdoor deployment.

Project Role

Environmental protection of sensor nodes

15. Complete Hardware Specification

Hardware	Main Function	Purpose in Project
Heltec WiFi LoRa 32 V3	Controller + LoRa	Controls sensor node and wireless communication
LoRa	Wireless communication	Transfers sensor data between nodes/gateway

Hardware	Main Function	Purpose in Project
MPU6050	Tilt + motion	Detects ground inclination/deformation
ADXL345	Acceleration/vibration	Measures quantitative vibration patterns
SW-420	Basic vibration trigger	Optional low-cost vibration detection
Linear displacement sensor	Measures linear movement	Detects ground displacement
Crack sensor	Crack opening detection	Detects early surface cracking
18650 Li-ion	Energy storage	Powers sensor node
Solar panel	Energy generation	Supports long-term autonomous operation
Charging/protection circuit	Battery management	Safely manages rechargeable power
ESP32 Deep Sleep	Power management	Extends battery life
Raspberry Pi	Gateway/edge computer	Collects, stores and processes node data
Buzzer/LED	Warning	Provides local alert
Weather-resistant enclosure	Protection	Protects electronics in outdoor conditions

16. Overall Hardware Working

The complete hardware operation can be represented as:

GROUND SURFACE

NODE 1 NODE 2 NODE 3

17. Hardware-to-Problem Mapping

The strongest part of your report should be showing **why each hardware component exists**.

PS Requirement	Hardware Solution
Detect abnormal ground tilt	MPU6050
Detect unusual vibration	ADXL345
Detect relative ground movement	Linear displacement sensor
Detect early cracks	Crack sensor
Continuous distributed monitoring	Multiple ESP32 sensor nodes
Long-range communication	LoRa
Wireless surface network	Multiple LoRa nodes + gateway
Low-cost implementation	ESP32 + low-cost sensors
Long battery operation	18650 + deep sleep
Renewable energy	Solar panel
Offline capability	Raspberry Pi local storage
Edge processing	Raspberry Pi
Local warning	Buzzer/LED
Outdoor deployment	Weather-resistant enclosure
AI prediction	Sensor data supplied to AI platform
GIS risk visualization	Gateway/cloud data integration

Recommended final hardware stack

For your team's SIH submission, I would present the hardware as **three layers**:

Layer 1 — Sensor Node

Heltec ESP32 LoRa V3 + MPU6050 + ADXL345 + displacement sensor + crack sensor + battery + solar

Layer 2 — Communication & Edge Gateway

LoRa network + Raspberry Pi

Layer 3 — Intelligence & Warning

Raspberry Pi → AI/ML → GIS → Dashboard → Warning system

This makes your hardware section look much more professional than simply listing components.

One important point: don't claim that these low-cost sensors alone can *certify* or definitively predict a mine collapse. In your SIH report, describe them as a **prototype for detecting deformation/anomalies and generating risk indications**, with thresholds and field deployment requiring calibration and validation by mining/geotechnical experts.